



POCKETPILOT AI

PHONE CONTROL

DEVELOPER TOOLS

See it. Say it. Fix it.

A phone-first, local-first developer assistant that turns visible failures into human-approved, test-verified fixes.

iQOO Hackathon 2026 | Team Zero Day 404

DEVELOPER ASSISTANT



Get into a reviewed patch and a real
the phone in your hand.

Repository stays on your laptop

LIVE WORKFLOW

ERROR

ROOT CAUSE FOUND

Missing value dereferenced

The repository lookup can return **None** accessed.

```
+ if user is None:  
+   return "Unknown"
```

✓ **FIX VERIFIED**
2 / 2 REAL TESTS PASSED

HUMAN APPROVAL REQUIRED

Debugging breaks when the failure is outside the IDE

```
Traceback (most recent call last):  
  File "user_service.py", line 18, in get_user_name  
    return users[user_id]["name"]  
KeyError: 42
```

01 The error is visible

A terminal, another machine, or a borrowed device shows the failure.

02 Context is fragmented

Copying traces, finding files, and validating a repair becomes manual work.

03 Trust is the real bottleneck

Developers need evidence, a readable diff, explicit approval, and rollback.

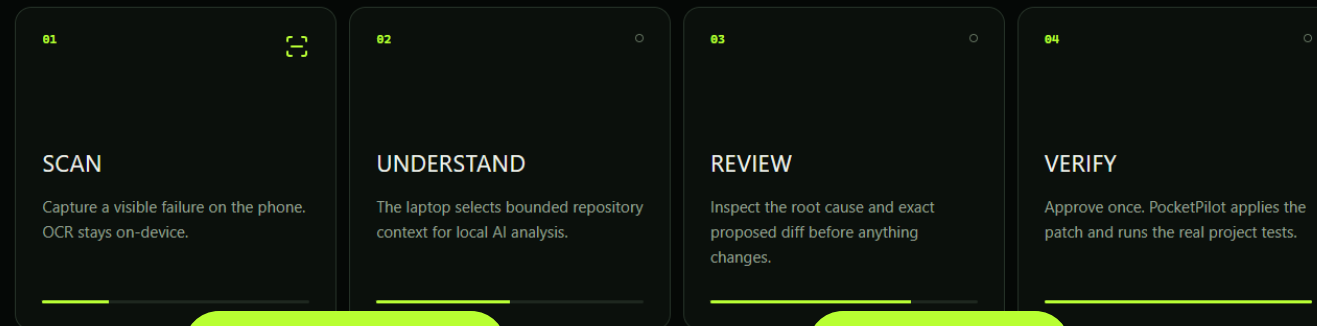
PocketPilot keeps the developer in control while moving the workflow to the phone.

One loop from visible failure to verified outcome

🔗 INTERACTIVE PRODUCT TOUR

One controlled path from failure to proof.

This tour illustrates the verified PocketPilot workflow. It is not connected to a live repository.



SCAN

Camera and OCR

UNDERSTAND

Local model + repo

REVIEW

Root cause + diff

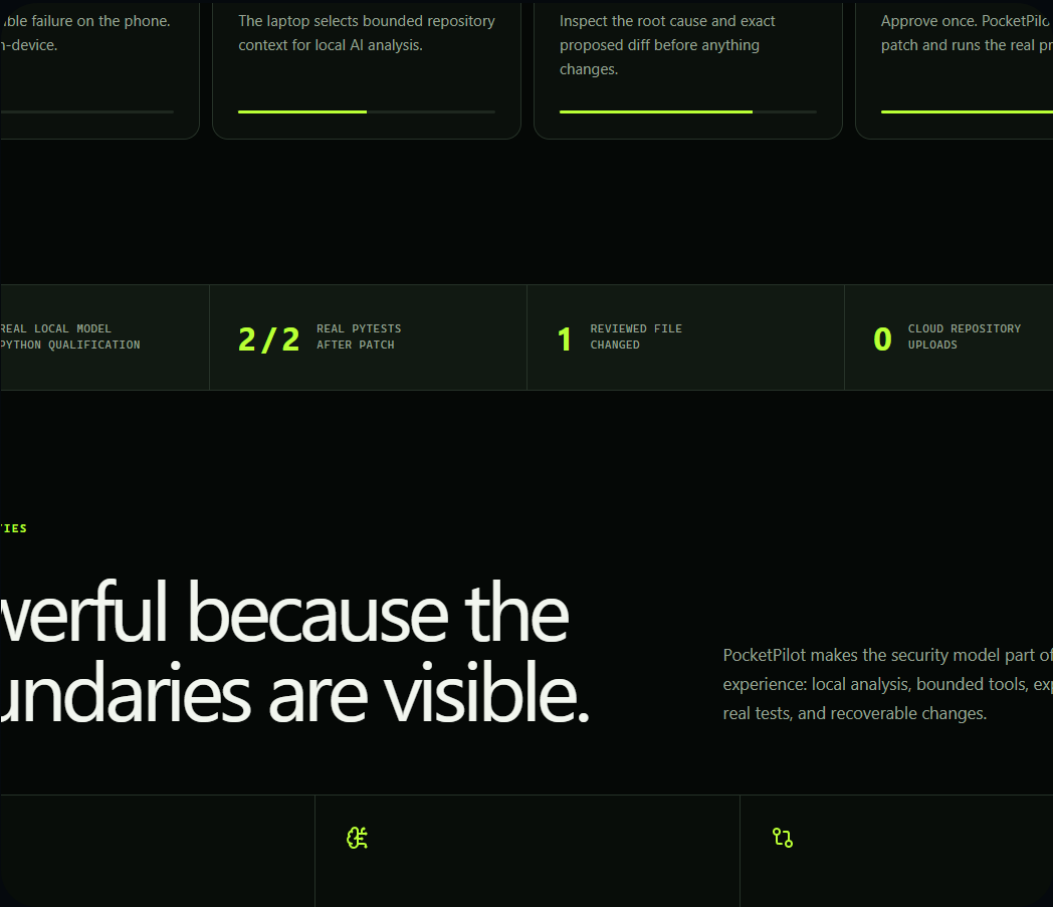
VERIFY

Tests + rollback

The phone becomes the safe control surface

CAPTURE	Camera scan and on-device OCR
COMMAND	Closed-vocabulary Android voice actions
REVIEW	Root cause, evidence, bounded source, and diff
CONTROL	Approve, reject, inspect, verify, and undo
FEEDBACK	Status, latency, speech, and progress animation

The phone never owns arbitrary project paths. It sends bounded intent to an authenticated laptop agent.



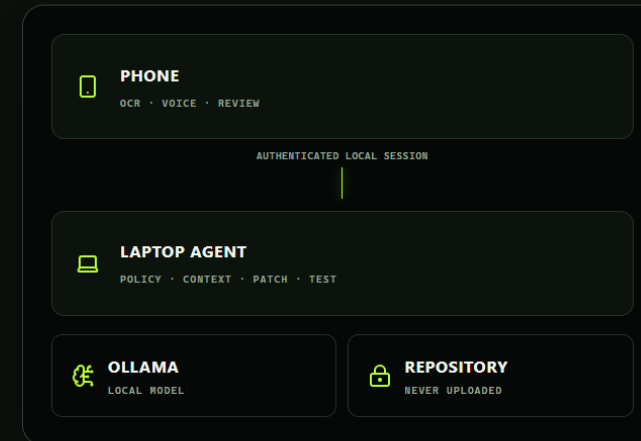
The phone directs. The laptop owns.

— LOCAL-FIRST ARCHITECTURE

The phone directs. The laptop owns.

The mobile app sends reviewed intent to an authenticated local agent. That agent alone owns repository paths, bounded context selection, patch application, verification, and rollback.

- ✓ Short-lived pairing and device session
- ✓ Registered projects and fixed validation commands
- ✓ Private local model endpoint



No source repository is uploaded to a cloud model. The selected workspace, validation command, and file boundaries remain laptop-owned.

Approval is a state, not a magic word

SCHEMA

Only known action shapes are accepted

EVIDENCE

Claims must point to bounded source context

PATH

Edits cannot escape the registered workspace

HASH

Stale proposals fail before write

DIFF

The exact patch is visible before approval

VERIFY

Only allowlisted validation commands run

ROLLBACK

Applied changes can be restored

ETY + PRIVACY

Approval is a state, not a magic word.

Nothing to confirm means nothing
cutes.

ing "yes" is accepted only when a specific
wable action is already awaiting
rmation.

02

Unsafe requests stay unsupported.

Arbitrary shell, filesystem, publishing, and
instruction-override commands are outside the
mobile command vocabulary.

03

A passing test is the finish line.

A patch is not called verified until the laptop r
the project's registered validation command
successfully.

Human approval stays mandatory before any file changes.

Evidence, not theatre

2 / 2**DEMO TESTS PASSED**

Python missing-user repair

7 / 7**REAL OLLAMA CYCLES**

Python scenarios

130**BACKEND TESTS PASSED**

4 environment skips

149**JAVASCRIPT TESTS PASSED**

Desktop, mobile, shared

21 / 21**EXPO DOCTOR CHECKS**

Native project health

1**PHYSICAL IQOO DEVICE**

Full apply, verify, undo

Also verified with WAN unavailable while the phone and laptop remained connected over the local network.

A real mobile, backend, protocol, and local-AI system

01

ANDROID APP

React Native + Expo

Camera, on-device OCR, speech service, review, approval, progress, undo

02

LAPTOP AGENT

FastAPI + SQLite

Pairing, sessions, repository context, transactional patching, verification

03

LOCAL MODEL

Ollama + qwen3-coder:30b

Root-cause analysis and structured repair proposals without cloud repo upload

04

TRUST PROTOCOL

Authenticated WebSocket

Bounded commands, replay resistance, evidence checks, state-aware approval

Registered demos cover Python, Java, and React/TypeScript scenarios.

Built for usefulness, originality, execution, and presentation

- | | | |
|-----|--------------|--|
| 30% | USEFULNESS | Shortens a fragmented debugging workflow while keeping the user in control. |
| 25% | ORIGINALITY | A phone-first approval and evidence surface for local developer automation. |
| 20% | EXECUTION | Working mobile app, authenticated bridge, local model, patching, tests, and rollback. |
| 15% | PRESENTATION | A single clear story with visible proof at every stage. |
| 10% | OFFICE KIT | Use the supplied kit for screen mirror, clipboard, file transfer, and remote control during the event. |

Office Kit is used as an official cross-device productivity tool. No unsupported vendor API integration is claimed.

READY FOR PHASE 1

From visible failure to verified outcome.

Scan the failure. Understand it locally. Review the exact change. Approve it deliberately. Verify it with real tests.

LIVE PRODUCT TOUR

pocketpilot-ai.sleek-pearl-0098.chatgpt.site

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o that
evidence,
atre.

INPUT

Real terminal failure

Camera or reviewed text

ANALYSIS

Local 30B coding model

Bounded repository context

CHANGE

Exact diff before approval

Risk and expected effect shown

PROOF

Project tests + undo

Execution result returned to phone